

SEDAR 8: Southeastern Spiny Lobster, Caribbean Spiny Lobster, & Caribbean Yellowtail Snapper

Southeastern Atlantic Spiny Lobster

Data Workshop

- Work to develop an active program for a juvenile tuning index
- Develop a greater understanding of the interaction between lobsters and traps
- Develop research partnerships with the fishery
- Try to reestablish an onboard fishing vessel monitoring program
- Increase understanding of lobster disease
- Continue to understand growth
- Develop future assessments that take into account the role males play in determining fecundity.

Assessment Workshop

- Tuning Indices: geographically robust adult and juvenile monitoring programs that could provide tuning indices that can be connected to each other and the fishery.
- Growth: lack of growth data from larger (>100 mm CL) lobsters

Review Workshop

- Data from the commercial fishery
 - * Re-establish a commercial fishery observer program (described above).
Fishery-independent indices of abundance
 - * Standardize existing data sets that may be used for juvenile and legal-sized indices of abundance
 - * Design new monitoring programs to collect systematic, consistent, and statistically rigorous data.
- Improved growth information
 - * Tagging projects should be initiated to obtain growth-rate data from larger (CL >100 mm) lobsters

- * Activity may need to be focused in areas of reduced exploitation (such as the Tortugas) to allow capture of these larger individuals in appreciable numbers
- * Reconcile growth information from Lipofuscin and tagging data
- Modeling
 - * Conduct Monte Carlo simulations to test F20% and F30% threshold and target reference points against various performance criteria. The stock assessment workshop for the stock should develop various scenarios covering a range of hypotheses concerning recruitment and changes in gear selectivity, as well as suitable performance indicators, including catch and measures of SSB. Risks in the performance indicators associated with applying the threshold and target should be generated in future assessments.
- Fishing pressure has decreased in the Keys because (i) there are less traps as a result of the Trap Certificate Program, (ii) recent efforts to curtail a rapidly expanding illegal dive fishery, (iii) the loss of dock space and subsequent selling out as gentrification continues at an increasing rate, (iv) the loss of suitable crew as a direct consequence of the increasing cost of living in the Keys.
- Fishermen are very willing to sit down with scientists to devise long-term observer/sampling programs that enmesh with operational activity and satisfy crucial needs for data.

Caribbean Spiny Lobster

Data Workshop

- Commercial Statistics
 - * Estimate landings based on complete catch report database after corrections to landings database are made and after reporting years 1986/1987 to 1992/1993 are entered.
 - * Recalculate expanded landings based on new lists of licensed fishers.
 - * Table final analyses of commercial bio-statistical data (size-frequency, catch composition, CPUE) until all the field sampling data has been completely entered and checked for errors and both US, Virgin Island and NMFS, SEFSC staff have signed off on corrections.
 - * Avoid repetitive analyses on incomplete information. Use only complete data sets in stock assessment analysis. A solid foundation will then be established for the analysis of other species to be included in future assessments.

- * Immediate changes in the catch report forms are not recommended. The fishing community in the U.S. Virgin Island is reluctant to provide any additional information, unless they see their data of approximately 30 years reflected in the management decisions.
- If the assessment proceeds, assumptions about the data should be clearly identified.
- Provide feedback to the fishing community after stock assessment analyses are performed, in order to reassure them that the information they provide is valuable and necessary to manage their resources.
- Caribbean Fishery Management Council staff present at the SEDAR8 Data workshop, recommended to conduct stock assessments with the information available at the moment to support management decisions. Proper consideration of uncertainty and acknowledgment of missing data was recommended.

Assessment Workshop

- Fishery-Independent Sampling:
 - * Increase the fishery independent sampling effort in the US Caribbean.; diversify regions sampled; cooperative sampling design and implementation between the fishermen and scientists; those species deemed important to the local fishing economy should be given sampling priority.
 - * Relatively good knowledge of habitat distributions and of habitat usage by various species/life stages provides a valuable opportunity to explore the power of habitat-based spatial models in this region.
- Fishery-Independent Monitoring of Spiny Lobster:
 - * Develop fishery independent sampling program specific to Caribbean spiny lobster.
 - * Visual surveys could be used in the Virgin Islands and in Puerto Rico to collect additional size and abundance information on the spiny lobster resource.
 - * Mark recapture techniques could be attempted to estimate abundance and learn more about the movements and habitat preferences of spiny lobster.

Review Workshop

- Improve and complete historical data on relative abundance indices and catch
- Fishery-independent monitoring

- * The Panel identified an apparent inconsistency between the assessment model assumptions of recruitment as a direct function of spawning stock. This appeared important enough to warrant two recommendations:
 - 1) build additional flexibility into the models to allow time-varying recruitment (or at least recruitment dynamics);
 - and,
 - 2) seek to establish a fishery-independent index of recruitment, which is deemed to be crucial.
- * The panel recommends considering the method used for the SA-GOM lobster assessment: placing a series of post-larval collectors in appropriate areas and consistent sampling their catch.
- * It is necessary to develop and implement sampling program(s) specific to both pre-recruit and adult Caribbean spiny lobsters
- * It is crucial to increase sampling effort in the US Caribbean.
- * There will be benefit in further diversifying the regions sampled to include equal coverage of areas frequently fished
- * Visual surveys for size structure, abundance, and YPR could provide useful time-series of data
- Revise the trip interview program (TIP) database exhaustively
 - * Completing the historical data set would be valuable
 - * Revitalizing TIP sampling in the US Virgin Islands would have many benefits, not just for the Caribbean spiny lobster stock
 - * Effort should be directed at key species, generating trip-target information, and obtaining needed detail
 - * Length distribution of the catch
- Commercial: Complete incorporation of non-digitized data for the US Virgin Islands (TIP). Recover historical length data for Puerto Rico and the US Virgin Islands from other studies prior to the TIP.
- Recreational: Determine length distributions

- Conduct studies to understand the ecology of early juveniles (25 mm carapace length)
 - * Habitat use needs to be understood better
 - * More needs to be known about settlement habitat
 - * Information on movements and migrations needs to be sought
 - * Clarity of the mortality rates needs to be sought
- Spatially explicit studies
 - * Identify spawning areas and sources of recruits
 - * Build/acquire habitat maps to identify stratification for research designs
 - * Combine habitat maps with density counts and habitat models to provide population estimates
 - * Develop a GIS map of spiny lobster landings throughout the geographic range of the stock, producing catch distributions
- Mark-recapture techniques
 - * Such studies could hone knowledge of abundance
 - * The techniques could provide additional information on movements and migrations
 - * Habitat preferences would be better understood
- Stock structure
 - * Stock structure is important in assessments, and genetics offers hope to improve knowledge
- Future assessments
 - * These should explore further use of length structure and density from closed areas as reference points
 - * Assessments need to be repeated when significant quantities of previously unavailable historical data have become available
 - * Alternative stock assumptions need to be considered during assessment: That of a wider Caribbean stock, That of the stock of the US Caribbean and neighboring islands

- * The use of nominal CPUE should be considered in future assessments
- * The modeling approach needs to be modified to produce a model that would support the observed data. Within the model, the recruitment parameter r should be allowed to increase over the second part of the time-series, perhaps moving beyond the standard modeling software currently used. Of the above, the Panel places the highest priority on the following, understanding the need to maximize the likelihood of generating an acceptable assessment of the stock in the near future:
 - * Develop/strengthen fishery-independent data collection
 - * Incorporate historical data into existing data sets
 - * Utilize refined models (better to identify viable hypotheses)

Caribbean Yellowtail Snapper

Data Workshop

Life History

- tagging studies of adult yellowtail snapper to obtain data on large-scale movements.
- evaluate maturation (size and spatial variation) and growth and fecundity
- preparation of general regional-wide GIS maps of landings

Commercial Statistics

- Complete data entry and clean-up task of fisher landings reports for reporting years 1986/1987 through reporting years 1992/1993) within 2-3 months, prior to the SEDAR8-Assessment Workshop. This task is currently being carried out by the US Virgin Islands, DFW;
- Estimate landings based on complete catch report database after corrections to landings database are made and after reporting years 1986/1987 through 1992/1993 are entered;
- Recalculate expanded landings based on new lists of licensed fishers;
- Staff of the US Virgin Islands, DFW suggested that analyses of commercial bio-statistical data (size-frequency, catch-composition, CPUE) should be put on hold until all the field sampling data has been completely entered and checked for errors and both US, Virgin Island and NMFS, SEFSC staff have signed off on corrections;

- Avoid repetitive analyses on incomplete information. Use only complete data sets in stock assessment analysis. A solid foundation will then be established for the analysis of other species to be included in future assessments;
- If assessments proceed with incomplete databases, assumptions about the data should be clearly identified and formally documented;
- Immediate changes in the fisher landings report forms are not recommended. The fishing community in the U.S. Virgin Islands is reluctant to provide any additional information, unless they see their data of approximately 30 years reflected in the management decisions;
- Provide feedback to the fishing community after stock assessment analyses are performed, in order to reassure them that the information they provide is valuable and necessary to manage their resources; and
- CFMC and NMFS, SEFSC staff present at the SEDAR8 Data workshop, recommended to conduct stock assessments with the information currently available to support management decisions. Proper consideration of uncertainty and documentation of missing or possibly inaccurate data was emphasized.

Overall workshop recommendations

- Continue the updating and data correction checks ongoing for the US Virgin Islands commercial landings and Biostatistical data bases.
- Continue the data correction checks ongoing with the Puerto Rico commercial landings and bio-statistical data bases.
- Continue the analyses related to partitioning of US Virgin Islands bulk landings data into species groupings after the missing bio-statistical samples have been entered, proofed and agreed on by both US Virgin Islands DFW staff and NMFS, TIP staff.
- Work toward developing a species specific commercial landings sales ticket in the US Virgin Islands commercial fisheries.
- Work towards research to obtain bio-statistical samples in the US Virgin
- Islands and especially to improve much needed sampling in St. Thomas/St. John. Fisheries.
- Implement hard part biological sampling in US Virgin Island and Puerto Rico.
- Work towards identifying the primary information needs regarding improving

- the ongoing fishery independent sampling initiatives for yellowtail snapper populations in the Caribbean.

Assessment Workshop

- Increase the fishery independent sampling effort in the U.S. Caribbean. Cooperative sampling design and implementation between the fishermen and scientists is strongly encouraged. If every species captured cannot be completely sampled, then those species deemed to be important to the local fishing economy or those species considered representative of relevant habitat types should be given sampling priority. A list of commercially important species to the region can be obtained from the Caribbean Fishery Management Council.
- The ideal survey would utilize hook and line and traps as the primary sampling gears in order to maintain consistency with those surveys that have been completed in the past.
- Visual surveys could be used in the Virgin Islands and in Puerto Rico to collect additional size and abundance information on the reef fish resource.
- Mark recapture techniques could be used to estimate abundance and learn more about the movements and habitat preferences of yellowtail snapper.
- The relatively good knowledge of habitat distributions and of habitat usage by various species/life stages provides a valuable opportunity to explore the power of habitat based spatial models in this region.

Review Workshop

- Fishery-independent data
 - * A new independent sampling regime to target yellowtail snapper more effectively should be created, because current methods do not allow temporal or spatial coverage.
 - * Visual surveys can provide useful fishery-independent data. The methods would, however, vary, based on the depth of the insular shelf.
 - * The output of other existing studies (NOAA and non-NOAA) should be examined to see if alternative fishery-independent sampling already exists.
- Life history data
 - * Fecundity data should be collected
 - * Maturity data should be collected
 - * Growth information should be collected

- * The parameter natural mortality needs investigation on the basis of better data
- Catch data
 - * Recreational catches need to be sampled and quantified better
 - * Information on trip species targeting is needed
 - * Information on the location of catches is sometimes not good, and should be
 - * improved
 - * Identification of species in the snapper complex in the US Virgin Islands is crucial to future assessments
 - * Historical data from the US Virgin Islands need to be collected from fishermen, if they exist
 - * Port samplers need to modify their schedules to target yellowtail snapper landings, and to sample sizes of the species need to increase
 - * TIP sampling in the US Virgin Islands needs to be revitalized
- Age and length frequency data
 - * These are needed from all commercial catches
 - * These are urgently required from recreational catches
 - * Fishery-independent surveys can provide these crucial data
- Genetic / otolith microchemistry studies
 - * Stock structure is important in assessments, and genetics and otolith microchemistry offer hope to unravel it in future
- Spatially explicit studies
 - * Identification of spawning areas and the source of recruits is important
 - * Construction of habitat maps will help identify stratification for research designs
 - * Combination of habitat maps with fish counts and habitat models will aid in providing population estimates
 - * Development of a GIS map of yellowtail snapper landings throughout the species' geographical range could help in the production of a distribution map of catches

- Mark-recapture studies
 - * This could help identify movements and migrations
 - * Fishing mortality estimates could be derived
 - * Population estimates would be enhanced with such studies
 - * Such studies could help solve the perplexing question of stock structure Of the above, the Panel places the highest priority on the following, understanding the need to maximize the likelihood of generating an acceptable assessment of the stock in the near future:
 - * The carrying out of fishery-independent surveys
- Collection of more catch data, including specifically the recreational fishery
- The collection of age and length data from commercial and recreational catches and from fishery-independent surveys
- Continue the updating and data correction checks ongoing for the US Virgin Islands commercial landings and Biostatistical data bases.
- Continue the data correction checks ongoing with the Puerto Rico commercial landings and bio-statistical data bases.
- Continue the analyses related to partitioning of US Virgin Islands bulk landings data into species groupings after the missing bio-statistical samples have been entered, proofed and agreed on by both US Virgin Islands DFW staff and NMFS, TIP staff.
- Work toward developing a species specific commercial landings sales ticket in the US Virgin Islands commercial fisheries.
- Work towards research to obtain bio-statistical samples in the US Virgin Islands and especially to improve much needed sampling in St. Thomas/St. John. Fisheries.
- Implement hard part biological sampling in US Virgin Island sand Puerto Rico.
- Work towards identifying the primary information needs regarding improving the ongoing fishery independent sampling initiatives for yellowtail snapper populations in the Caribbean.

Review Workshop Procedural Suggestions for SEDAR

- There is a strong need for enhanced communication, specifically to stakeholders, about what SEDAR is trying to achieve in terms of management.
- To date, there has not been full acceptance from all, and this is put down at least partially to the lack of education and training of certain key parties about the process. Their cooperation is essential if SEDAR is to succeed in its objectives.
- An advanced plan of what species is to be handled when is essential for all those who need and wish to be involved in the process.
- There is need for a (web-based) Glossary of Terms used.
- Continuity of personnel in the workshops is crucial to ensuring both acceptance and enhanced understanding.
- Dissemination of the information created and the results in terms of management action are not always perceived by stakeholders to have been achieved, so it was felt that Councils should make greater effort in this regard, at all levels of the process.
- Several participants, both technical and representing fishermen, felt that greater effort should be made to maximize the time for preparation of data series, assessments, and review material. The Panel shied away from suggesting a deadline for receipt of material prior to each workshop, realizing that the very nature of some data would always make collection to the last possible moment necessary, but stressed that late receipt could easily lead to delayed or less informative assessments of stock status.
- As mentioned several times elsewhere in this report, strong cases were made for incorporating fishermen's knowledge better into the assessment and management process.
- The Review Panel requires the presence of scientists who have not been involved in the Data and/or Assessment Workshops. This may not be a preferred requirement for the participating stakeholders. Stakeholders would clearly benefit and be better able to participate fully in the review process if they had been present throughout all meetings. The Councils could maximize meeting this recommendation by considering paying stipends to participating stakeholders to compensate them for lost earnings.
- There was strong feeling that the anticipated changed representation on the Review Panel may not be most appropriate for the SEDAR area. While understanding and wholeheartedly endorsing the need for independent peer review, a strong case could be made for Panel representation to include stakeholders, biologists knowledgeable about the species, and stock assessment scientists who were not involved in the immediate assessment. It was felt unlikely that such people would be able to participate in the

discussions at the current enthusiastic level unless they were formally accepted as members of the Panel.

- Allied to the above and notwithstanding what was ultimately decided on the make-up of the Panel, there was unanimity that the independence of the Review Panel chair (currently appointed by the CIE) was paramount and matched well the objective of independence.
- Given the volume of documentation associated with such reviews and the shortage of time often available to assimilate it, the Review Panel and other participants stressed the need for a clear executive summary to be provided for all substantive documents being addressed. Further, there was a call for a succinct table of model parameters (estimated and observed) to be provided for each assessment along with, if appropriate, a table of management options (e.g. a decision table) and the risks associated with them.

Review Workshop Stakeholder recommendations

- The need for robust education of fishermen and other stakeholders is acknowledged. Such education should be of a two-way nature and would potentially lead to an enhancement of their trust in the assessment and management process, especially if they were to become involved in research program design.
- The fact that most of the product in the yellowtail snapper fishery is sold retail and that there are no fish houses (at least in the US Virgin Islands) makes any meaningful future stock assessment in the region extremely dependent on cooperation with the local fishermen.
- A paucity of recent socio-economic information continues to hinder the development of integrated biological, economic, and social assessments.
- Partnerships with organizations such as NGOs, which are often staffed by highly qualified people and are perhaps also less constrained by political influence, can mobilize extra resources in meeting some of the research objectives.
- Biological and habitat/ecosystem research information is as important in the assessment process as catch data.
- Over the past 35+ years of fishing, yellowtail snapper abundance has remained stable.
- Detailed data (information) on yellowtail snapper catch are lacking for US Virgin Islands commercial landings. The lack of this type of data has introduced uncertainty into the determination of stock status. Therefore, collection of detailed catch information there is suggested as a top research priority.

Recommendations of the CIE contractors

- ensure the provision of a large-scale locator map in the meeting room (for those not familiar with the geography or sampling areas).
- ensure that membership of Panels for future SEDAR Review Workshops preserves independence of any involvement in assessment of the stocks being addressed, in terms of both Chair and Panel (the latter to retain participation if possible by several US scientists not involved in the assessment).
- Yellowtail Snapper: In terms of future research and monitoring, much needs to be done, but to maximize the likelihood of generating an acceptable assessment of the stock in the near future, the highest priority should be on:
 - carrying out fishery-independent surveys;
 - collecting more catch data, including specifically the recreational fishery; and
 - collecting age and length data from commercial and recreational catches and from fishery-independent surveys
- Caribbean Spiny Lobster: priority for future research and monitoring was given to• developing/strengthening fishery-independent data collection;
- incorporating historical data into existing data sets; and
- utilizing refined models (better to identify viable hypotheses).
- Generally, the standardization procedure for the Caribbean yellowtail and spiny lobster abundance indices was well conducted and, based upon what was presented, the analyses appear to be sound. However, some improvements in the approach were recommended.
- Statistical criteria should not be the sole basis for determining terms in the GLM, but terms need to refer to some theoretical justification.
- Year interaction terms to remove random effects should be avoided if possible, as they could make the standardized index worse.
- Some factors would be better treated as covariates rather than factors, thereby reducing the number of parameters.
- The analysis needs to explore alternative treatments for missing data, rather than having a missing data category.